

Automatic Generation of Multimodal 4D Effects for Immersive Video Watching Experiences

Seoyong Nam*
Hanyang University
South Korea
namsyong0508@hanyang.ac.kr

Minho Chung
Hanyang University
South Korea
chungminho@hanyang.ac.kr

Haerim Kim
Hanyang University
South Korea
haelim010@hanyang.ac.kr

Eunchae Kim
Hanyang University
South Korea
dmsco0223@hanyang.ac.kr

Taehyeon Kim
Hanyang University
South Korea
kthday@hanyang.ac.kr

Yongjae Yoo
Hanyang University
South Korea
yongjaeyoo@hanyang.ac.kr

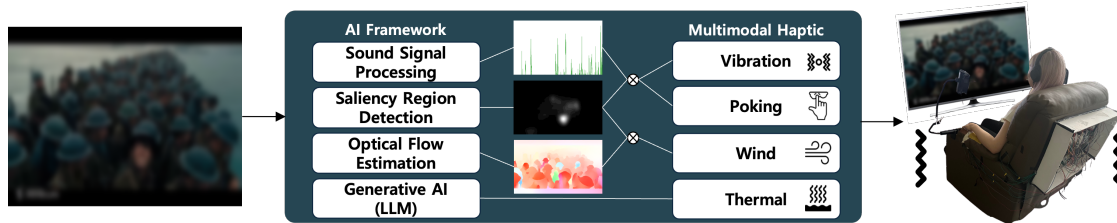


Figure 1: Overview of the system.

Abstract

The recent trend of watching content using over-the-top (OTT) services pushes the 4D movie industry to seek a way of transformation. To address the issue, this paper suggests an AI-driven automatic 4D effects generation algorithm applied to a low-cost comfort chair. The system extracts multiple features using psychoacoustic analysis, saliency detection, optical flow, and an LLM-based thermal effect synthesis, and maps them into various sensory displays such as vibration, heat, wind, and poking automatically. To evaluate the system, a user study with 21 participants across seven film genres was conducted. The results showed that 1) there was a general improvement with 4D effects in terms of immersion, concentration, and expressiveness, and 2) multisensory effects were particularly useful in action and fantasy movie scenes. The suggested system could be directly used in current general video-on-demand services.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; • **Information systems** → **Multimedia content creation**.

Keywords

4D films, haptics, multimodal, AI-driven haptic effects generation

ACM Reference Format:

Seoyong Nam, Minho Chung, Haerim Kim, Eunchae Kim, Taehyeon Kim, and Yongjae Yoo. 2024. Automatic Generation of Multimodal 4D Effects for Immersive Video Watching Experiences. In *SIGGRAPH Asia 2024 Technical Communications (SA Technical Communications '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3681758.3698021>

1 Introduction

4D movies, movies with extra sensory effects, have become a major trend in the movie industry in the last decade. With the enabling technology of 4D motion chairs for movie theaters and amusement parks, film distributors worldwide dived into the new market with their movies with tailor-made 4D effects. The audiences *ride* on the chairs during their watching, and the motion effects provide better user experiences [Danieau et al. 2014]. However, such a trend has changed drastically recently—people prefer watching movies at home, instead of theater, with personalized content using over-the-top (OTT) services such as Netflix or Disney+, which provide millions of video content.

There are two major streams of research direction towards 4D effects. One is automation, and the other one is exploration of different sensory modalities. First, automation is considered mainly due to the high cost of production. Many haptic designers spend a long time designing 4D effects. For example, a team of haptics experts spend two weeks per movie to create 4D effects for a piece of movie on average [Lee et al. 2015]. To reduce such efforts and the cost, automatic generation algorithms have been studied [Abreu et al. 2023; Lee et al. 2015, 2023]. Namely, Lee et al. [Lee et al. 2015] organized the taxonomy of motion effects by scene characteristics, e.g., camera or object moving, slow or fast moving, and so on.



This work is licensed under a Creative Commons Attribution International 4.0 License.

SA Technical Communications '24, December 03–06, 2024, Tokyo, Japan
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1140-4/24/12
<https://doi.org/10.1145/3681758.3698021>

This line of studies analyzed the movie scenes and automatically generated motion effects.

The other stream, the exploration of multimodality, has two main purposes: improving expressiveness and seeking low-cost alternatives. Li et al., studied vibrotactile 4D effect on a home theater environments [Li et al. 2021], for the alternative method to support 4D effects with movies during the COVID-19 pandemic. They used saliency maps and psychoacoustic features to determine the location and strength of the output vibration. Lin et al. used various low-cost IoT devices to allow designers to create 4D effects [Lin et al. 2018] effectively while designers were watching the movie. The study by Zhou et al. used various AI algorithms, including computer vision and audio analysis, to extract features that could be converted into motions, water jets, lighting, and vibrations [Zhou et al. 2018]. The evaluation of their study focused on assessing the appropriateness of the automatically generated effects. Studies on multisensory, multimodal stimuli presented with movie clips were performed, which showed that such stimuli provide richer and more immersive experiences [Jeong et al. 2021; Yun et al. 2023].

Considering the current trends of at-home, OTT-based media viewing experience, there are growing opportunities for 4D movie services on top of those media providers. In this sense, we designed and proposed an AI-based automated system for generating multimodal haptic effects for home theater environments. The algorithm analyzes the video and extracts audio and visual features, and generates thermal effects suggested by an LLM model (GPT-4). Such features and effects are converted to multisensory stimuli consisting of thermal, wind, poking, and vibrotactile effects, and presented by the actuators implanted in a comfort chair.

2 System Implementation

We implemented both hardware and software. As hardware, we assembled a comfort chair equipped with vibration, wind, thermal, and poking haptic displays. For software, we designed an AI framework that orchestrates several AI algorithms. The framework extracts audio and visual features from the video input and then converts them into haptic rendering signals. Finally, the signals were transmitted to the hardware, and 4D effects were presented to the audience.

2.1 Audio Feature Extraction and Rendering

We started from Li et al.'s study [Li et al. 2021] for the audio feature extraction method, given that the study also targets the home theater environments. We calculated psychoacoustical features of loudness, sharpness, and booming from audio input, which indicates perceived volume, and the portion of high-pitch and low-pitch components, respectively. We tried to elaborate the algorithm used in [Li et al. 2021], by introducing poking stimuli and adaptive thresholding. In [Li et al. 2021], all the features were mapped to vibrations with fixed threshold values. Instead, we mapped sharpness to poking and loudness and booming to vibrations. For every frame, we calculated the threshold (ϕ) values of loudness, sharpness, and booming using the moving average (μ) values of the features and standard deviation (σ) of adjacent 60 frames. (See eq. 1) If the feature value exceeds the moving average plus 1.645 times of standard deviation, the corresponding haptic effects will be generated. As a

result, poking and vibrations, were generated by the salient parts of the sound input.

$$\phi_{feat}[i] = \mu(feats[i : i - 59]) + 1.645\sigma(feats[i : i - 59]) \quad (1)$$

$$v[i] = \begin{cases} v[i] & \text{if } v[i] \geq \phi_{loudness}[i] \text{ or } v[i] \geq \phi_{booming}[i] \\ 0 & \text{o.w.} \end{cases} \quad (2)$$

$$p[i] = \begin{cases} p[i] & \text{if } p[i] \geq \phi_{sharpness}[i] \\ 0 & \text{o.w.} \end{cases} \quad (3)$$

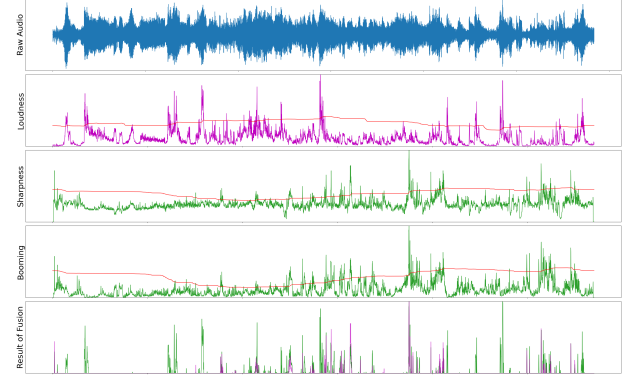


Figure 2: Sound analysis results of Elemental (66 s; Table 1). (Top to bottom) Raw audio, loudness, sharpness, booming, and haptic rendering signal output. The orange line indicates the adaptive threshold value over the scene.

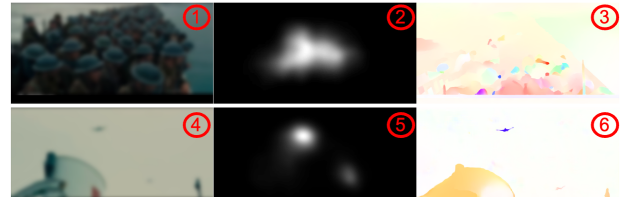


Figure 3: (Scenes from Dunkirk) In ①, a crowd of soldiers moving, has a large salient region (②) but a small optical flow (③). In contrast, air raid of fighter jets (④)–smaller salient region but large flow (⑤, ⑥). Output $w[i]$ is larger in ④ instead of ①–strong wind will be presented in the air raid scene.

2.2 Visual Feature Extraction

As visual features, we selected saliency map, optical flow, and LLM-based thermal description generation. First, to determine the location of vibration effects, we derived a saliency map, using TASED-NET [Min and Corso 2019] model. The extracted saliency map at frame i , $S[i]$ was divided into a 2x4 grid and mapped to each vibration actuator to provide the vibration effects. For optical flow at frame i to its adjacent frame $i-1$, $F[i, i-1]$ calculation, we used Flowformer [Huang et al. 2022] model, which showed the best performance at that stage. To detect the flow effectively, we first applied the saliency map and removed non-salient regions from the optical flow output to determine which objects and movements

are *salient*. The results were mapped into the airflow direction and strength to render wind effects.

Considering that the camera is not fixed but follows objects in most cases, we used the difference of salient regions between two features—the vanilla salient region and the optical flow’s salient region—to determine the wind parameters. This means that we calculated the mean squared distance (MSD) between the saliency map $S[i]$ and the saliency map of optical flow $F[i, i - 1]$ in pixel units. (Note that all pixels in $S[i]$, $F[i, i - 1]$ are binary.) Then, the wind output value $w[i]$ was calculated as MSD divided by the size of the saliency region, $\sum_j S[i]$.

$$\text{MSD}(F[i, i - 1], S[i]) = \frac{1}{N} \sum_{j=1}^N (F[i, i - 1]_j - S[i]_j)^2 \quad (4)$$

$$w[i] = \frac{\text{MSD}(F[i, i - 1], S[i])}{\sum_{j=1}^N S[i]_j} \quad (5)$$

Lastly, to avoid “*too weak, too frequent*” wind output, we thresholded the $w[i]$ below the grand mean of the MSD, T . The overall process is described in Fig. 3.

$$T = \frac{1}{N} \sum_{i=1}^N \text{MSD}(F[i, i - 1], S[i]) \quad (6)$$

$$w[i] = \begin{cases} w[i] & \text{if } w[i] \geq T \\ 0 & \text{o.w} \end{cases} \quad (7)$$

Lastly, we utilized GPT-4’s video captioning for thermal rendering, to exploit the GPT model’s ability to “*understands the context*.” We instructed GPT-4 to generate the scene’s appropriate temperature rendering into seven levels: [very cold, cold, cool, normal, warm, hot, very hot]. Fig. 4 shows a few examples.

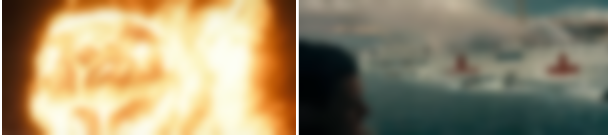


Figure 4: Temperature level inference examples: a fire dragon scene (Harry Potter; left) was labeled to “very hot”, and swimming scene (Dunkirk; right) was “cool”.

2.3 Haptic Rendering on The Hardware

As a haptic hardware platform, we used a comfort chair equipped with the following output components: eight ERM motors driven by L298N motor drivers, two solenoids with relay circuits for poking effects, two Peltier modules with large heatsinks and fans controlled by MD10C DC drivers for thermal effects, two fans for wind effects. An Arduino Mega 2560 was used for communication and hardware control. A desktop PC was used for playing the movie clips and transmitting the signal simultaneously.

All frame-wise rendering features $v[i]$, $p[i]$, $w[i]$, $t[i]$ for i -th frame from the AI framework were stored. Considering the PWM range of the Arduino, $v[i]$ is normalized to a range of 80–180 (0.0 mapped to 80, 1.0 to 180). $w[i]$ was multiplied by 255 (note that $w[i]$ below T were wiped out; minimum value of $w[i]$ signal was $T \times 255$). Thermal rendering is categorized into 7 levels (0 to 6),

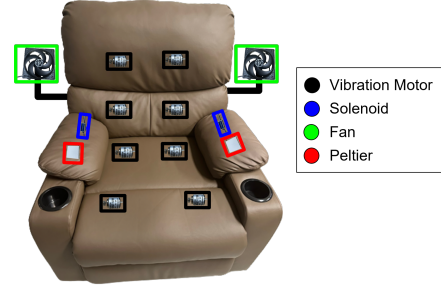


Figure 5: Multimodal Haptic Hardware Configuration.

Table 1: Movie clips used in the user experiment

Movie	Len.	Genre	Still Shot	Description
Elemental (2023)	66 s	Animation		A fire character riding a motorcycle rushes to save his family before the dam collapses.
Dark knight (2008)	90 s	Action		The Joker blows up an entire hospital building.
Dunkirk (2017)	90 s	War		A plane drops bombs while the sound of cannons echoes nearby during a naval battle.
Harry Potter (2005)	120 s	Fantasy		Two wizards are engaged in a magical duel
Notting Hill (1999)	135 s	Romance		A man and woman confess their love during a press conference.
Insidious (2012)	60 s	Horror		A trailer introduces a ghost incident that occurred to a family.
Interstellar (2014)	95 s	Sci-fi		A man entering a black hole, experiencing multiple collisions and tremors.

and poking is represented in a binary format (on and off). While the rendering software is running, the control parameters of each actuator are sent to the Arduino via serial communication. The Arduino parses and controls the actuators accordingly.

3 User Experiment

Using the implemented system, we conducted a user study designed to evaluate the experience of multimodal 4D effects in a simulated home-theater environment.

3.1 Setup and Procedure

21 university students volunteered for the study, who had no prior experience in video design and had used motion chairs no more than twice (12 males, 9 females, average 23.4 years old). Participants were instructed to sit in the 4D chair and wear wireless noise canceling headphones. The experiment consisted of two sessions. We initially asked about the participants’ anticipation in the first session. Then, we provided three 4D effects across three conditions: N (no haptic effect), V (vibration only), and M (multimodal) for seven movie clips (in Table 1), resulting in 21 trials per participant. The presenting order of the three conditions and seven movie clips was counterbalanced. In each trial, participants were asked to watch the movie clip and feel the 4D effects. Then, they fill out a survey that asks for immersion, concentration on the movie, and expressiveness

Table 2: Results of the user study. Boldfaced: $p < .05$, bold-asterisk: $p < .01$. Cell color coding: Tukey's post-hoc test results, significant differences coded with different colors.

Movie	Item	No haptics	Vibration	Multimodal
Elemental	Immersion*	47.85	68.66	76.90
	Concentration*	49.28	67.33	71.42
	Expressiveness*	34.52	66.57	79.76
Dark knight	Immersion*	47.14	63.09	74.04
	Concentration	45.23	63.33	67.38
	Expressiveness*	30.23	62.38	73.57
Dunkirk	Immersion*	48.57	72.38	83.80
	Concentration*	47.14	73.57	84.04
	Expressiveness*	30.00	70.47	85.95
Harry Potter	Immersion	44.76	70.09	85.19
	Concentration*	44.04	67.61	81.42
	Expressiveness*	30.71	68.52	86.57
Notting Hill	Immersion	61.76	52.52	48.57
	Concentration	56.19	50.00	50.28
	Expressiveness	41.90	44.19	46.52
Insidious	Immersion	55.00	62.14	68.09
	Concentration	51.19	61.90	66.71
	Expressiveness*	38.09	59.61	69.66
Interstellar	Immersion*	52.14	74.28	77.85
	Concentration	54.04	69.95	73.80
	Expressiveness*	37.28	73.90	78.71

on a 100-point scale. After finishing the trials, the participants evaluated overall system satisfaction and provided free-form feedback. The procedure took about 70 mins per participant.

3.2 Results and Discussion

Table 2 summarizes the overall results. We could observe that the existence of 4D effects generally improves the watching experiences in terms of immersion, concentration, and expressiveness in general. Especially, for Dunkirk and Harry Potter, the auto-generated multimodal effects were evaluated as even better than unimodal ones. It suggests that our autogenerated multimodal 4D effects from audiovisual elements could be effective methods for 4D content. In particular, war, battle, and fantasy magic scenes showed significant improvements in the participant's watching experience. These scenes contain explosions, battles, salient actions with and audiovisual effects, and the auto-generated multisensory 4D effects seem to provide sensory redundancy, yielding rich sensory experiences.

However, as seen in Table 2, such automatic extraction and conversion seem to rely on movie genres and contents. The 4D effects did not improve the watching experiences of Notting Hill and Insidious. These clips contain some lyrical, calm scenes or heavily rely on the story context. For such types of movies, alternative mechanisms for 4D effects would be needed, given that dialog and the expressions of the main characters are pivotal.

Another notable trend is the dominant role of vibrations in our 4D effects. The user experience scores of the V conditions are similar to those of M conditions. The reason would be twofold: in the first section of anticipation, most participants evaluated the vibration's contributions at least 30 percent across all genres (average of 41.42%). This might be partially related to 1) the familiarity with the modality (whole-body vibration), and 2) the area of contact with the actuators (eight across the body). In the post-experimental survey, participants answered that they thought that the system generally worked and presented 4D effects as they expected.

However, considering the energy and sensory information coded and delivered to each modality, wind, heat, and poking also showed their potential. In fantasy and battle scenes, thermal and poking, wind effects contributed to better immersion and expressiveness scores. Participants' comments in the post-experimental surveys echo: "Multimodal experiences fit in battle scenes well," "Multimodal approach was more effective in urgent scenes." Still, further exploration is required for improved use cases of such modalities.

4 Conclusions and Future Work

In this study, we implemented and evaluated a system that automatically extracts the multimodal 4D effects of vibration, poking, wind, and heat. The orchestration of multiple AI models could extract the features in the movie clips and convert them to adequate haptic stimuli. The user study showed that 1) multimodal 4D effects of our system improved the watching experience and 2) such 4D effects benefit active scenes such as war and fantasy scenes, 3) further investigation is needed for some story-driven genres. Our future works include expanding the algorithms for scents and light effects and seeking generative AI's potential for haptic features.

Acknowledgments

This work was supported by Hanyang University (HY-2023-2575), Excellence in SW Program No. 2024-0-00058, and a research grant No. RS-2023-00226263 from Korean government.

References

- Raphael Abreu, Douglas Mattos, Joel Santos, George Guinea, and Débora C Muchaluat-Saade. 2023. Semi-automatic mulsemmedia authoring analysis from the user's perspective. In *Proc. ACM Multimedia*. 249–256.
- Fabien Danieau, Julien Fleureau, Philippe Guillotel, Nicolas Mollet, Marc Christie, and Anatole Lécuyer. 2014. Toward Haptic Cinematography: Enhancing Movie Experiences with Camera-Based Haptic Effects. *IEEE MultiMedia* 21, 2 (2014), 11–21.
- Zhaoyang Huang, Xiaoyu Shi, Chao Zhang, Qiang Wang, Ka Chun Cheung, Hongwei Qin, Jifeng Dai, and Hongsheng Li. 2022. Flowformer: A transformer architecture for optical flow. In *Proc. ECCV*. Springer, 668–685.
- Dong Yeong Jeong, Sung H Han, Seungmoon Choi, Dawoon Jeong, and Kimin Kwon. 2021. Investigating perceived emotions and affects of a scene, and the user satisfaction with motion effects in 4D movies. *Int. J. Ind. Ergon.* 85 (2021), 103173.
- Jaebong Lee, Bohyung Han, and Seungmoon Choi. 2015. Motion effects synthesis for 4D films. *IEEE Trans. Visual Comput. Graphics* 22, 10 (2015), 2300–2314.
- Jiwan Lee, Sung H Han, and Seungmoon Choi. 2023. Sensory cue integration of visual and vestibular stimuli: a case study for 4D rides. *Virtual Reality* 27, 3 (2023), 1671–1683.
- Yaxuan Li, Yongjae Yoo, Antoine Weill-Duflos, and Jeremy Cooperstock. 2021. Towards context-aware automatic haptic effect generation for home theatre environments. In *Proc. VRST*. 1–11.
- Yi-Bing Lin, Ming-Ta Yang, and Yun-Wei Lin. 2018. Low-cost four-dimensional experience theater using home appliances. *IEEE Trans. Multimedia* 21, 5 (2018), 1161–1168.
- Kyle Min and Jason J Corso. 2019. Tased-net: Temporally-aggregating spatial encoder-decoder network for video saliency detection. In *Proc. ICCV*. 2394–2403.
- Gyeore Yun, Minjae Mun, Jungeun Lee, Dong-Geun Kim, Hong Z Tan, and Seungmoon Choi. 2023. Generating real-time, selective, and multimodal haptic effects from sound for gaming experience enhancement. In *Proc. ACM CHI*. 1–17.
- Yuhao Zhou, Makarand Tapaswi, and Sanja Fidler. 2018. Now you shake me: Towards automatic 4D cinema. In *Proc. IEEE CVPR*. 7425–7434.